

Smart Traffic Lights Using Local AI

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Abstract—Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights generally follow fixed timers and cannot easily react when traffic conditions change suddenly. A local artificial intelligence based system can process traffic information near the road and make faster decisions. This paper presents a conceptual smart traffic-light system using local AI and camera-based traffic information to improve signal control.

Index Terms—Smart Traffic Lights, Local AI, Artificial Intelligence, Traffic Management, Machine Vision, Traffic Signal Control

I. INTRODUCTION

Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. They are simple and inexpensive, but they cannot easily react when one road suddenly becomes crowded.

Recent systems use cameras and artificial intelligence (AI) to estimate how many vehicles are present and then adjust the lights. Instead of changing lights only according to a clock, the system can use information collected from the road.

Research reviews also show growing interest in AI-based traffic signal control and its possible benefits for changing traffic conditions [1].

II. LITERATURE REVIEW

Cloud-based AI systems send camera information to a central server for processing. This can provide strong computing power, but it depends on an internet connection and can introduce communication delay.

Local smart-camera systems move some processing closer to the road. This can reduce the amount of information that must be sent over the internet and may allow faster decisions.

Machine-vision research has also explored using camera-based vehicle information to adapt traffic-light timing [2].

III. PROBLEM STATEMENT

Traditional traffic-light systems commonly use fixed timing mechanisms. Such systems may not respond efficiently to sudden changes in traffic density.

An important emergency problem is the difficulty of automatically creating a clear route for an emergency vehicle such as an ambulance. Therefore, there is a need for a smart traffic system that can detect traffic conditions and coordinate nearby signals to provide a faster response.

IV. PROPOSED METHODOLOGY

The proposed system uses street cameras to capture traffic conditions. The captured information is processed using a local computer equipped with an AI-based traffic detection mechanism.

The system follows these general steps:

- 1) Capture traffic information using a camera.
- 2) Process the captured information locally.
- 3) Estimate the traffic density.
- 4) Determine an appropriate signal duration.
- 5) Send the decision to the traffic-light controller.
- 6) Update the traffic signal.

Local processing can reduce the amount of information that needs to be transmitted to a remote server.

V. SYSTEM ARCHITECTURE

The proposed architecture consists of street cameras, smart pole computers, a city cloud server, and traffic lights.

The basic flow is:

Street Cameras → Smart Pole Computers → City Cloud Server → Traffic Lights

Selected information can be forwarded to the cloud server for storage and wider analysis, while local processing can be used for quick traffic decisions.

A. Image Insertion

Figure 1 shows an example of the proposed system architecture.

VI. SYSTEM FLOWCHART

The flowchart represents the basic operation of the proposed traffic-light system.

VII. MATHEMATICAL MODEL

A simple traffic-density measure can be represented as:

$$D = \frac{N_v}{A} \quad (1)$$

where D represents traffic density, N_v represents the number of detected vehicles, and A represents the observed road area.

The system can use the estimated density to determine the required green-light duration.

For example:

traffic_system.png

Fig. 1. Proposed Smart Traffic Light System Architecture

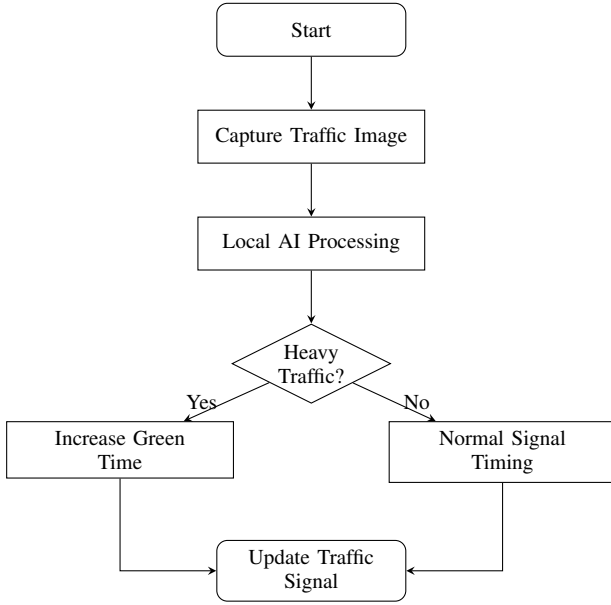


Fig. 2. Flowchart of the proposed traffic-light system

$$T_g = T_{\min} + kD \quad (2)$$

where T_g is the green-light duration, $T_{\min}$ is the minimum green-light duration, and k is a control parameter.

VIII. TRAFFIC DATA REPRESENTATION

Traffic information from different road directions can be represented using a matrix.

$$A = \begin{bmatrix} N_1 & N_2 \\ N_3 & N_4 \end{bmatrix} \quad (3)$$

where N_1 , N_2 , N_3 , and N_4 represent the number of vehicles detected in different traffic lanes or directions.

A larger matrix can also be used when multiple lanes are monitored:

$$B = \begin{bmatrix} 10 & 15 & 20 \\ 8 & 12 & 18 \\ 5 & 10 & 14 \end{bmatrix} \quad (4)$$

IX. SYSTEM COMPARISON

Table I provides a simple comparison of different traffic-light approaches.

TABLE I
COMPARISON OF TRAFFIC-LIGHT SYSTEMS

System	Decision Time	Internet	Cost
Basic Timed Lights	30 s	Zero	\$200
Cloud AI System	2 s	High	\$2500
Local Smart Camera	0.5 s	Low	\$1200

X. ALGORITHM

The basic operation of local traffic detection can be represented using Algorithm 1.

Algorithm 1 Local Traffic Detection

- 1: Capture traffic image
- 2: Process image using local AI
- 3: Count detected vehicles
- 4: **if** vehicle count is high **then**
- 5: Increase green-light duration
- 6: **else**
- 7: Use normal signal timing
- 8: **end if**
- 9: Update traffic signal

XI. IMPLEMENTATION EXAMPLE

A simple programming example for calculating traffic density is shown below.

Listing 1. Simple Traffic Density Calculation

```

1 #include <stdio.h>
2
3 int main()
4 {
5     int vehicles = 25;
6     int roadArea = 100;
7     float density;
8
9     density = (float)vehicles / roadArea;
10
11     printf("Traffic Density = %.2f\n", density);
12
13     return 0;
14 }

```

XII. RESULTS AND DISCUSSION

The proposed local AI architecture is intended to respond to traffic conditions using information collected from the road. Local processing can reduce dependence on continuous communication with a remote server.

The system can potentially use traffic density information to adjust signal timing. The actual performance of a deployed system would depend on camera quality, AI model accuracy, computing hardware, traffic conditions, and signal-controller design.

XIII. RESEARCH GAP

Existing traffic systems can work well for regular daily traffic, especially when traffic patterns are predictable. However, an important emergency problem remains: many systems do not have a built-in way to automatically clear a path for a speeding ambulance.

This creates a need for a smart traffic system that can detect an emergency vehicle and quickly coordinate nearby signals to create a clear route.

XIV. ADVANTAGES

The proposed approach has the following potential advantages:

- Faster local decision-making.
- Reduced dependence on internet connectivity.
- Reduced amount of information sent to cloud servers.
- Ability to respond to changing traffic conditions.
- Potential support for emergency-vehicle routing.

XV. LIMITATIONS

The effectiveness of the system depends on several factors, including:

- Camera quality.
- Accuracy of the AI model.
- Local computing capability.
- Weather and lighting conditions.
- Correct communication with traffic-light controllers.

XVI. FUTURE SCOPE

Future implementations can explore improved vehicle detection, emergency-vehicle recognition, adaptive signal coordination, and integration with larger traffic-management systems.

XVII. CONCLUSION

This paper presents a conceptual smart traffic-light system based on local AI and camera-based traffic information. The approach moves some processing closer to the road, which can support faster traffic decisions and reduce dependence on continuous cloud communication.

The proposed architecture can be further developed to detect emergency vehicles and coordinate nearby traffic signals to create a clear route.

REFERENCES

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